The edge exponent of the uniform doubling condition, II: the matching upper bound

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Abstract

In the companion note [*The edge exponent of the uniform doubling condition and an interpolation conjecture*] we proved that for i.i.d. uniform samples on the unit ball truncated by a single transversal half-space, the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] fails with probability $1-o(1)$ at every scaling factor below $c(\log N/N)^{2/(n+2)}$, and we conjectured the interpolation law: critical exponent $2/(2n-j)$ when the lowest transversal boundary stratum of the truncation regions has dimension j. Here we prove the matching upper bound: the condition holds with probability $1-1/N$ at scaling $\ell_N = O\big((\log N/N)^{2/(n+2)}\big)$. Hence the critical scaling for the half-space truncation of the ball is exactly $\Theta\big((\log N/N)^{2/(n+2)}\big)$, and the interpolation law is now proved at all three anchor points $j=n-1$, $j=n-2$ and $j=0$. The geometric heart is a uniform section bound for $G = B^n \cap \{x_1 \le 0\}$: at depth h below the extreme point of G in *any* direction, the level section has $(n-1)$-volume at least $c_n h^{n/2}$. Its proof splits into one half-section estimate for feasible directions and, for infeasible directions, two overlapping regimes about the angular scale $\sin\varphi \asymp \sqrt{h}$, each elementary.

1 Setting and result

Let $K = B^n$ be the closed unit ball, $n \ge 2$, μ the uniform probability measure on K, $T = \{X_1, \dots, X_N\}$ i.i.d. samples with empirical measure $\bar{\mu}_N$, and

$$H := \{x : \ x_1 \le 0\}, \qquad G := K \cap H.$$

Truncated slabs $S_w = \{x : |\langle x - s, u\rangle| \le \frac{w}{2}\} \cap H$ and the uniform doubling condition (UDC) with constants $(\kappa, \delta, \sigma, \beta, \ell)$ for the single truncation set $\{(e_1, 0)\}$ are as in [1, 2, 6]. In [6] we proved the lower bound: with probability $1-o(1)$ (for $n \ge 3$) the UDC requires $\ell \ge c(\log N/N)^{2/(n+2)}$. Here we complete the picture.

Theorem 1.1. *Let $n \ge 2$. With the constants $\kappa = 1 + 150$, $\sigma = 1$, $\delta = \delta(n) > 0$, $\beta = 2$ of the companion papers (here $\widetilde{R} = 1$ and the thickness radius is $r_0 = \frac{1}{2}$, since $B(-\frac{1}{2}e_1, \frac{1}{2}) \subseteq G$), there are $C(n)$ and $N_0(n, \kappa)$ such that for all $N \ge N_0$, with probability at least $1 - \frac{1}{N}$, the sample satisfies the UDC for the truncation set $\{(e_1, 0)\}$ with scaling factor*

$$\ell_N \ = \ C(n) \left(\frac{\log N}{N}\right)^{2/(n+2)}.$$

Consequently, by [6], the critical scaling for this truncation is exactly $\Theta\big((\log N/N)^{2/(n+2)}\big)$, and the interpolation exponent $\frac{2}{2n-j}$ of [6, Conjecture 1.2] is established at $j = n-2$, alongside the proven cases $j = n-1$ [3, 4] and $j = 0$ [3, 5].

By the proof skeleton of [2], as isolated in [3, Section 2], Theorem 1.1 reduces to a deterministic estimate: the doubling inequality holds at all scales $w \in [\ell_N, \beta)$ on the VC event provided every truncated slab of width w containing a sample point has widened population mass $\mu(S_{\kappa w}) \geq 32u_N$, where $u_N \asymp \log N/N$ is the VC rate; and this follows from a uniform slab-volume bound. We therefore prove:

Theorem 1.2 (Slab volume)**.** *There are $c(n) > 0$ and $L_0(n) > 0$ such that for every $x \in G$, every unit vector u, and every $0 < L \leq L_0$,*

$$\mathrm{vol}\Big(G \cap \big\{y : \ |\langle y - x, u\rangle| \leq L\big\}\Big) \ \geq \ c(n)\, L^{\frac{n+2}{2}}.$$

Granting Theorem 1.2, the proof of Theorem 1.1 is verbatim the assembly of [3, Proof of Theorem 1.2] (equivalently [4, Section 3]): the population dichotomy of [2] applies since G contains a $\frac{1}{2}$-ball; the VC relative deviations apply to the class of truncated slabs; a sample point $x \in S_w$ gives, via Theorem 1.2 applied with $L = \min\big(\frac{(\kappa-1)w}{2}, L_0\big)$ and the inclusion $\{|\langle y - x, u\rangle| \leq L\} \subseteq S'_{\kappa w}(u, s)$,

$$\mu(S_{\kappa w}) \ \geq \ \frac{c(n)}{\mathrm{vol}(B^n)} \min\Big(\tfrac{(\kappa-1)w}{2},\, L_0\Big)^{\frac{n+2}{2}} \ \geq \ 32\, u_N \qquad (w \geq \ell_N),$$

by the definition of ℓ_N and, for large w, the choice of N_0. The rest of this paper proves Theorem 1.2.

2 The section lemma

For a unit vector u write $\tau(x) := \langle x, u\rangle$, $\tau_{\max}(u) := \max_G \tau$, and

$$A_u(h) \ := \ \mathrm{vol}_{n-1}\Big(G \cap \{\tau(x) = \tau_{\max}(u) - h\}\Big).$$

Decompose $u = u_1 e_1 + u'$ with $u' \perp e_1$, and set $\sin\varphi := u_1$ when $u_1 > 0$.

Lemma 2.1 (Section lemma)**.** *There are $a(n) > 0$ and $h_0(n) \in (0, \frac{1}{4}]$ such that for every unit vector u and every $0 < h \leq h_0$,*

$$A_u(h) \ \geq \ a(n)\, h^{\,n/2}.$$

The three exponent regimes behind the proof: for $u_1 \leq 0$ the bound holds at the better rate $h^{(n-1)/2}$; for $u_1 > 0$ the extreme sits on the edge (or face) and the rate $h^{n/2}$ is sharp.

Proof. Case I: $u_1 \leq 0$. Here the unconstrained maximizer of τ over K, namely $x = u$, lies in H, so $\tau_{\max} = 1$ with extreme point u, and the level set of K at depth h is the $(n-1)$-ball

$$D_h := K \cap \{\tau = 1 - h\}, \qquad \text{center } c_h = (1-h)u, \quad \text{radius } r_h = \sqrt{2h - h^2} \geq \sqrt{h}.$$

If $u = -e_1$, then $D_h = \{x_1 = 1 - h\} \cdot (-1)$ — precisely, the level set is $\{x \in K : x_1 = -(1-h)\} \subseteq H$, so $A_u(h) = \mathrm{vol}_{n-1}(D_h) = \omega_{n-1} r_h^{n-1}$. Otherwise e_1 is not parallel to u and

the trace of ∂H on the level hyperplane is an affine hyperplane of that hyperplane; the center satisfies $(c_h)_1 = (1-h)u_1 \le 0$, i.e. the center lies in the kept half, and a half-ball through the center is kept:

$$A_u(h) \;\ge\; \tfrac{1}{2}\,\omega_{n-1}\,r_h^{n-1} \;\ge\; \tfrac{\omega_{n-1}}{2}\,h^{\frac{n-1}{2}} \;\ge\; \tfrac{\omega_{n-1}}{2}\,h^{\frac{n}{2}},$$

using $h \le 1$.

Case II: $u_1 = \sin\varphi > 0$, $u' \neq 0$. Write $p := u'/|u'|$, so $|u'| = \cos\varphi$ and, for $x \in G$,

$$\tau(x) = \sin\varphi\, x_1 + \cos\varphi\,\langle x, p\rangle \;\le\; 0 + \cos\varphi\cdot|x| \;\le\; \cos\varphi,$$

with equality iff $x_1 = 0$, $x = p$: the extreme is the edge point p, and $\tau_{\max} = \cos\varphi$. Let

$$\Lambda(h) \;:=\; \mathrm{vol}\big(\{x \in G :\; \tau(x) \ge \cos\varphi - h\}\big) \;=\; \int_0^h A_u(t)\,dt.$$

We first show

$$\Lambda(h) \;\ge\; b(n)\,h^{\frac{n+2}{2}} \qquad (0 < h \le h_0), \tag{1}$$

and then convert (1) into the section bound.

Regime IIa: $\sin^2\varphi \ge h\cos\varphi$. Write $h' := h/\cos\varphi \le 1$ and note the regime means $h' \le \tan^2\varphi$. Parametrize $x = (-t, x_2, x'')$ in coordinates where $p = e_2$: the cap is $\{t \ge 0,\ |x| \le 1,\ x_2 \ge 1 - h' + t\tan\varphi\}$. Restrict to

$$0 \le t \le \frac{h'}{4\tan\varphi}, \qquad |x''|^2 \le \frac{h'}{2}.$$

Then $t\tan\varphi \le \frac{h'}{4}$ and $t^2 \le \frac{h'^2}{16\tan^2\varphi} \le \frac{h'}{16}$ by the regime condition, so, using $\sqrt{1-s} \ge 1 - s$,

$$\sqrt{1 - t^2 - |x''|^2} - \big(1 - h' + t\tan\varphi\big) \;\ge\; h' - t\tan\varphi - t^2 - |x''|^2 \;\ge\; h'\Big(1 - \tfrac{1}{4} - \tfrac{1}{16} - \tfrac{1}{2}\Big) \;=\; \tfrac{3h'}{16},$$

i.e. the x_2-fibre has length at least $\frac{3h'}{16}$ on the whole restricted (t, x'')-box. Hence

$$\Lambda(h) \;\ge\; \frac{3h'}{16}\cdot\frac{h'}{4\tan\varphi}\cdot\omega_{n-2}\Big(\frac{h'}{2}\Big)^{\frac{n-2}{2}} \;=\; b_1(n)\,\frac{h'^{\frac{n+2}{2}}}{\tan\varphi} \;=\; b_1(n)\,h^{\frac{n+2}{2}}\cdot\frac{\cos^{-\frac{n}{2}}\varphi}{\sin\varphi} \;\ge\; b_1(n)\,h^{\frac{n+2}{2}},$$

since $\cos^{-n/2}\varphi \ge 1$ and $\sin\varphi \le 1$.

Regime IIb: $\sin^2\varphi \le h\cos\varphi$. Then $1 - \cos\varphi \le \sin^2\varphi \le h$, so with $\hat h := h + (1-\cos\varphi) \le 2h$ the cap contains $\big\{x \in K :\; \langle x, p\rangle \ge 1 - \frac{\hat h}{16},\ x_1 \le 0\big\}$: indeed for such x, $|x-p|^2 = 2(1 - \langle x, p\rangle) \le \frac{\hat h}{8}$ gives $|x_1| \le \sqrt{\hat h/8}$, whence

$$\tau(x) \;\ge\; -\sin\varphi\sqrt{\tfrac{\hat h}{8}} + \cos\varphi\Big(1 - \tfrac{\hat h}{16}\Big) \;\ge\; \cos\varphi - \Big(\sqrt{h\cos\varphi}\cdot\sqrt{\tfrac{2h}{8}} + \tfrac{2h}{16}\Big) \;\ge\; \cos\varphi - h,$$

using $\sin\varphi \le \sqrt{h\cos\varphi}$, $\hat h \le 2h$ and $\frac{h}{2} + \frac{h}{8} \le h$. The set $\{\langle x, p\rangle \ge 1 - \frac{\hat h}{16}\} \cap K$ is a spherical cap centered at p, and $p_1 = 0$, so the reflection $x_1 \mapsto -x_1$ maps it to itself and exactly half of it lies in $\{x_1 \le 0\}$. Hence

$$\Lambda(h) \;\ge\; \tfrac{1}{2}\,\mathrm{vol}\Big(\text{cap of depth } \tfrac{\hat h}{16}\Big) \;\ge\; b_2(n)\,h^{\frac{n+1}{2}} \;\ge\; b_2(n)\,h^{\frac{n+2}{2}},$$

by the standard two-sided cap-volume asymptotic and $h \le 1$. The two regimes cover all $\varphi \in (0, \frac{\pi}{2})$, proving (1).

From cap mass to sections. By the Brunn–Minkowski inequality, $A_u^{1/(n-1)}$ is concave on the range of τ over G, hence A_u is unimodal in the depth variable h. We first record a bulk anchor. The inner ball $B^- := B(-\frac{1}{2}e_1, \frac{1}{2}) \subseteq G$ has τ-center $m^- := \tau(-\frac{1}{2}e_1) = -\frac{1}{2}\sin\varphi$, and its sections at τ-distance at most $\frac{1}{4}$ from m^- have radius at least $\sqrt{\frac{1}{4} - \frac{1}{16}} = \frac{\sqrt{3}}{4}$. The corresponding depths are $h = \tau_{\max} - \lambda$ with $\lambda \in [m^- - \frac{1}{4}, m^- + \frac{1}{4}]$, i.e. $h \in [h^\dagger - \frac{1}{4}, h^\dagger + \frac{1}{4}]$ where $h^\dagger := \tau_{\max} - m^- = \cos\varphi + \frac{1}{2}\sin\varphi \in [\frac{1}{2}, \frac{9}{8}]$ for $\varphi \in (0, \frac{\pi}{2})$. In particular there is a depth $h^\dagger \ge \frac{1}{2} \ge h_0$ with $A_u(h^\dagger) \ge \omega_{n-1}\big(\frac{\sqrt{3}}{4}\big)^{n-1} =: a_1(n)$.

Now let $h \le h_0$ and let h^* be a maximizer of A_u. If $h \le h^*$, then A_u is nondecreasing on $[0, h]$ and

$$A_u(h) \;\ge\; \frac{1}{h}\int_0^h A_u \;=\; \frac{\Lambda(h)}{h} \;\ge\; b(n)\, h^{n/2}$$

by (1). If $h > h^*$, then A_u is nonincreasing on $[h^*, h^\dagger] \ni h$, so $A_u(h) \ge A_u(h^\dagger) \ge a_1(n) \ge a_1(n)h^{n/2}$. In both cases $A_u(h) \ge a(n)h^{n/2}$ with $a(n) := \min\{b(n), a_1(n)\}$.

Case II-degenerate: $u' = 0$, i.e. $u = e_1$. Then $\tau_{\max} = 0$, attained on the whole face $K \cap \{x_1 = 0\}$, and $A_u(h) = \mathrm{vol}_{n-1}(K \cap \{x_1 = -h\}) = \omega_{n-1}(1-h^2)^{\frac{n-1}{2}} \ge a(n)$ for $h \le \frac{1}{2}$. (The case $u = -e_1$ was covered in Case I.) □

3 The slab-volume bound

Proof of Theorem 1.2. Fix $x \in G$, u, and $L \le L_0 := \frac{h_0}{4}$. Let $\tau_{\max} := \tau_{\max}(u)$ and $\tau_{\min} := \min_G \tau = -\tau_{\max}(-u)$, and let $A := A_u$ denote the section profile, extended over the full range $[\tau_{\min}, \tau_{\max}]$, which has length at least 1 (the inner ball $B(-\frac{1}{2}e_1, \frac{1}{2})$ has τ-range of length 1). By Lemma 2.1 applied to u and to $-u$,

$$A(\tau_{\max} - a) \;\ge\; a(n)\Big(\tfrac{L}{4}\Big)^{n/2} =: A^*, \qquad A(\tau_{\min} + a') \;\ge\; A^*,$$

with anchor depths $a = a' = \frac{L}{4}$. By unimodality (Brunn–Minkowski), $A \ge A^*$ on the middle interval $M := [\tau_{\min} + \frac{L}{4},\ \tau_{\max} - \frac{L}{4}]$, whose complement in the range has length $\frac{L}{2}$. The slab's level interval $[\tau(x) - L, \tau(x) + L]$ meets the range in an interval of length at least L (the range has length $\ge 1 \ge 2L$ and contains $\tau(x)$), hence meets M in a set of length at least $\frac{L}{2}$. Therefore

$$\mathrm{vol}\Big(G \cap \{|\langle y - x, u\rangle| \le L\}\Big) = \int_{\tau(x)-L}^{\tau(x)+L} A \;\ge\; \frac{L}{2}\, A^* \;=\; \frac{a(n)}{2 \cdot 4^{n/2}}\, L^{\frac{n+2}{2}}.$$

□

4 Conclusion

Theorem 1.1 follows, as explained in Section 1, by inserting Theorem 1.2 into the proof skeleton of [2, 3]. Together with [6], the critical scaling of the uniform doubling condition for the half-space truncation of the ball is $\Theta\big((\log N/N)^{2/(n+2)}\big)$, and the interpolation law $\ell_N \asymp (\log N/N)^{2/(2n-j)}$ is now proved at its three anchor strata: smooth ($j = n-1$), transversal edge ($j = n-2$), and conical point ($j = 0$). The remaining cases $1 \le j \le n-3$ — transversal strata of intermediate

dimension, produced by $n-1-j$ half-spaces in general position — should follow from the same two-step scheme: a section lemma with rate $h^{(2n-j-2)/2}$ via the regime split at the angular scale $\sqrt{h}$ in each transverse direction, and the multi-cap second moment of [4, 5, 6] for the lower bound. We leave the general case to future work.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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